Program	Bachelor of Technology (B.Tech)	**Semester - 5**
Type of Course	Core Courses	
Prerequisite	MACHINE LEARNING ALGORITHMS	
Rationale	1. Define the concepts of Artificial Intelligence and their uses. 2. Explain the importance of knowledge representation in AI systems. 3. Understand the basics of probabilistic reasoning, including probability and Bayes' theorem. 4. Understand the different stages of NLP, including syntactic and semantic analysis, discourse, and pragmatic processing. 5. Understand the different stages of NLP, including syntactic and semantic analysis, discourse, and pragmatic processing.	
Effective From A.Y.	2024-25	

Teaching Scheme (Contact Hours)				**Examination Scheme**				
Lecture	**Tutorial**	**Lab**	**Credit**	**Theory Marks**		**Practical Marks**		**Total Marks**
				T	**T**	**P**	**P**	
3	0	2	4	70	30	50	-	150

***SEE** - Semester End Examination, **T** - Internal Theory, **P** - Internal Practical*

Course Content

T - Teaching Hours | **W** - Weightage

Sr.	**Topics**	**T**	**W**
1	**Introduction to Artificial Intelligence and Intelligent Agents** Introduction to Artificial Intelligence, History and Evolution of AI, Applications of AI, Intelligent Agents and Agent, Environments, Characteristics of Intelligent Systems, AI Problem Formulation, Types of AI: Narrow AI, General AI, Super AI, AI Techniques and Approaches, Ethics and Challenges in AI, Real-world AI Applications: Healthcare, Finance, Robotics Autonomous Vehicles, Recommendation Systems	**12**	**15**
2	**Problem Solving and Search Techniques** Problem Solving using AI, State Space Representation, Uninformed Search Techniques: Breadth First Search (BFS), Depth First Search (DFS), Uniform Cost Search, Informed Search Techniques: Best First Search, A* Algorithm, Hill Climbing, Constraint Satisfaction Problems, Game Playing Techniques, Mini-Max Algorithm, Alpha-Beta Pruning	**12**	**20**
3	**Knowledge Representation and Reasoning** Knowledge Representation Techniques, Predicate Logic and Propositional Logic, Semantic Networks, Frames and , Scripts, Ontologies, Forward and Backward Chaining, Rule-Based Expert Systems, Reasoning Under Uncertainty, Bayesian Networks, Fuzzy Logic Systems	**12**	**20**
4	**Machine Learning and Neural Networks** Introduction to Machine Learning, Types of Learning: Supervised Learning, Unsupervised Learning, Reinforcement Learning, Classification and Regression, Decision Trees, K-Nearest Neighbors (KNN), Basics of Neural Networks, Perceptron Model, Deep Learning Overview, Applications of Machine Learning in AI	**12**	**25**
5	**Natural Language Processing and AI Applications** Introduction to Natural Language Processing (NLP), Text Processing and Tokenization, Syntactic and Semantic Analysis, Speech Recognition, Chatbots and Conversational AI, Computer Vision Basics, Robotics and Intelligent Automation, AI in IoT and Smart Systems, Generative AI and Large Language Models, Future Trends in Artificial Intelligence	**12**	**25**
	Total	**60**	**100**

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy

Level	Remembrance	Understanding	Application	Analyze	Evaluate
Weightage	20	15	15	20	20

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:	
CO1	Understand the concepts, techniques, and applications of Artificial Intelligence.
CO2	Apply search algorithms and problem-solving techniques to AI problems.
CO3	Analyze and represent knowledge using logical and probabilistic reasoning methods.
CO4	Implement basic machine learning and neural network models for intelligent systems.
CO5	Evaluate AI applications in NLP, robotics, computer vision, and emerging technologies.

CO PO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5
PO - 1	3	3	3	2	2
PO - 2	3	2	3	2	3
PO - 3	2	2	2	2	3
PO - 4	2	2	3	2	2
PO - 5	2	2	2	2	3
PO - 6	2	1	1	2	2
PO - 7	1	1	1	1	1
PO - 8	1	1	1	1	1
PO - 9	2	2	2	2	2
PO - 10	1	2	1	2	1
PO - 11	-	1	1	1	2
PO - 12	2	2	2	2	3

Reference Books

1.	**“Artificial Intelligence (TextBook)** By Elaine Rich And Kevin Knight (2nd Edition) | Tata Mcgraw-Hill
2.	**Artificial Intelligence: A Modern Approach** By Stuart Russel, Peter Norvig, | PHI

List of Practical	
1.	Write a program to implement Tic-Tac-Toe game problem.
2.	Write a program to implement BFS (for 8 puzzle problem or Water Jug problem or any AI search problem).
3.	Write a program to implement DFS (for 8 puzzle problem or Water Jug problem or any AI search problem)
4.	Write a program to implement Single Player Game (Using any Heuristic Function)
5.	Write a program to Implement A* Algorithm.
6.	Write a program to implement mini-max algorithm for any game development.
7.	Assume given a set of facts of the form father(name1,name2) (name1 is the father of name2).
8.	Define a predicate brother(X,Y) which holds iff X and Y are brothers. Define a predicate cousin(X,Y) which holds iff X and Y are cousins. Define a predicate grandson(X,Y) which holds iff X is a grandson of Y. Define a predicate descendent(X,Y) which holds iff X is a descendent of Y. Consider the following genealogical tree: father(a,b). father(a,c). father(b,d). father(b,e). father(c,f). Say which answers, and in which order, are generated by your definitions for the following queries in Prolog:
9.	Write a program to solve Tower of Hanoi problem using Prolog.
10.	Write a program to solve N-Queens problem using Prolog.
11.	Write a program to solve 8 puzzle problem using Prolog.
12.	Write a program to solve travelling salesman problem using Prolog.